

Plant Care App - Architecture Documentation

Project Overview

React-based plant care management application with photo upload, care tracking, and bulk operations.

Technology Stack

- **Frontend:** React 18+ with Hooks
- **Styling:** Tailwind CSS (responsive design)
- **State Management:** React useState hooks + Custom hooks
- **Data Storage:** Local state (no database yet)
- **Photo Storage:** Base64 encoding in browser memory

File Structure

```
plant-care-app/  
├── public/  
│   └── images/  
│       └── port-photo.png (background image)  
├── src/  
│   ├── components/  
│   │   ├── PlantCard.js  
│   │   ├── AddPlantForm.js  
│   │   ├── EditPlantForm.js  
│   │   ├── CareTimeline.js  
│   │   ├── BulkCarePanel.js  
│   │   ├── PlantDetailView.js  
│   │   └── PhotoUpload.js  
│   ├── hooks/  
│   │   ├── usePlants.js  
│   │   ├── useCareActivities.js  
│   │   └── useBulkCare.js  
│   ├── utils/  
│   │   └── careUtils.js  
│   ├── App.js  
│   └── PlantManager.js (main orchestrator ~300 lines)
```

Data Models

Plant Object

```
javascript

{
  id: "unique-string",
  name: "Plant Name",
  species: "Portulacaria Afra",
  acquisitionDate: "2024-01-15",
  source: "Local nursery",
  price: 15.99,
  notes: "Description",
  photos: [photoObject, ...],
  coverPhotoId: "photo-id-string"
}
```

Photo Object

```
javascript

{
  id: "unique-string",
  url: "base64-encoded-image",
  fileName: "image.jpg",
  uploadDate: "2024-01-15T10:30:00Z",
  caption: "Optional description"
}
```

Care Activity Object

```
javascript

{
  id: "unique-string",
  plantId: "plant-id",
  activityType: "watering|fertilizing|pest-control|repotting|structural-work",
  date: "2024-01-15T10:30:00Z",
  notes: "Optional details"
}
```

Component Architecture

PlantManager.js (Main Orchestrator)

- **Role:** Central state coordinator
- **Responsibilities:** UI state management, form handling, navigation
- **Hooks Used:** usePlants, useCareActivities, useBulkCare
- **Key Functions:** handleAddPlant, handleEditPlant, handlePhotoUpload

Custom Hooks

usePlants.js

- **State:** plants array
- **Functions:** addPlant, updatePlant, deletePlant, addPhotoToPlant, deletePhotoFromPlant, setCoverPhoto

useCareActivities.js

- **State:** careActivities array
- **Functions:** logCareActivity, getRecentCareActivities, getPlantActivities

useBulkCare.js

- **State:** bulkCareMode, selectedPlantIds
- **Functions:** toggleBulkCare, toggleSelectPlant, selectAllPlants, applyBulkCare

UI Components

PlantCard.js

- **Props:** plant, bulkCareMode, isSelected, onToggleSelect, onViewPlant, onStartEdit, onDeletePlant, onLogCare, onOpenPhotoUpload, recentActivities
- **Features:** Cover photo display, care action buttons, management buttons
- **Layout:** Responsive card with conditional elements

PlantDetailView.js

- **Props:** plant, plantActivities, onBackToCollection, onStartEdit, onLogCare, onOpenPhotoUpload, onSetCoverPhoto, onDeletePhoto
- **Features:** Integrated photo gallery, complete care history, large care action buttons
- **Layout:** Full-page plant view with photo management

AddPlantForm.js / EditPlantForm.js

- **Props:** Form data, onChange handlers, onSubmit, onCancel
- **Features:** Validation, species dropdown, price formatting
- **Layout:** Modal-style forms with consistent styling

CareTimeline.js

- **Props:** activities, plants
- **Features:** Chronological activity feed, plant name resolution
- **Layout:** Scrollable timeline with activity cards

BulkCarePanel.js

- **Props:** selectedPlantIds, onSelectAll, onClearAll, onBulkCare
- **Features:** Bulk selection controls, limited care actions
- **Layout:** Horizontal action panel

PhotoUpload.js

- **Props:** plantId, onPhotoUpload, onClose
- **Features:** Multi-file selection, preview grid, caption input, base64 conversion
- **Layout:** Modal with drag-drop area and preview grid

Utility Functions (careUtils.js)

- `formatCareDate(dateString)`: Relative time formatting ("2h ago")
- `getActivityDisplay(activityType)`: Emoji and name mapping
- `formatPrice(price)`: Currency formatting with thousands separators

User Workflows

Plant Management

1. Add Plant → AddPlantForm → PlantCard appears
2. Edit Plant → EditPlantForm → PlantCard updates
3. View Plant → PlantDetailView → Full information + photo gallery
4. Delete Plant → Confirmation → PlantCard removed

Photo Management

1. Upload Photos → PhotoUpload modal → Base64 conversion → Storage in plant.photos
2. View Gallery → PlantDetailView shows integrated gallery
3. Set Cover Photo → Updates plant.coverPhotoId
4. Delete Photo → Removes from plant.photos array

Care Activity Workflow

1. Individual Care → PlantCard care buttons → logCareActivity
2. Bulk Care → Select plants → BulkCarePanel → Apply to multiple
3. View History → CareTimeline or PlantDetailView

Current Limitations

- No data persistence (refresh loses data)
- Base64 photos consume browser memory
- No user authentication
- No cloud storage
- No real-time sync
- No care reminders yet

Next Development Priorities

1. Supabase integration for data persistence
2. Care reminder system
3. Photo storage optimization
4. User authentication
5. Instagram export functionality

Mobile Responsiveness

- Grid layouts: `grid-cols-1 md:grid-cols-2 lg:grid-cols-3`
- Button sizing: Responsive text and padding
- Header: Stacked layout on mobile
- Photo gallery: Single column on mobile
- Forms: Full-width inputs on mobile

State Management Patterns

- UI state: Managed in PlantManager
- Business logic: Isolated in custom hooks
- Form data: Controlled components with useState
- Photo handling: File → Base64 → Local state storage